PRODUCT DESIGN · WHEAT PROCUREMENT · UNITED FLOUR MILL, THAILAND

Grain Desk

A single source of truth for a flour mill's wheat buying — built around the one number that decides it: what this cargo costs against what everyone else paid.

Design proposal, not a build Source: 71-minute walkthrough with Pavin Sethbhakdi, 4 Oct 2026

Timestamps cite the recording

WHY THE UNITS MATTER

Every number has to be legible in dollars per cargo

ONE CARGO	AT ~$350/T	A $5/T MOVE	WHEAT AS SHARE OF MILL COST
55,000 t	**$19.25M**	**$275,000**	**80%**
US wheat, Supramax with gear, PNW to Koh Sichang	Financed per vessel, 9–10 vessels a year	"It can increase like that within a few days"	Procurement is most of the income statement

Pavin does not think in dollars per tonne. He converts, every time, into what a move costs across a 55,000-tonne cargo — and then into the baht-per-kilo flour price he has to quote a customer next quarter. A product that shows `$358/t` and stops there is making him do the arithmetic that matters.

Design rule: no price appears in one unit. Always $/t, $/cargo, and where the chain reaches that far, ฿/kg of flour.

WHAT THE RECORDING SHOWED THAT THE NOTES DID NOT

He has already built four fifths of this in Excel

The summary notes described fragmentation and a wish list. The screen-shares show something more useful: a working system, assembled by hand, whose shape tells us exactly what to build. Three corrections and one omission change the design.

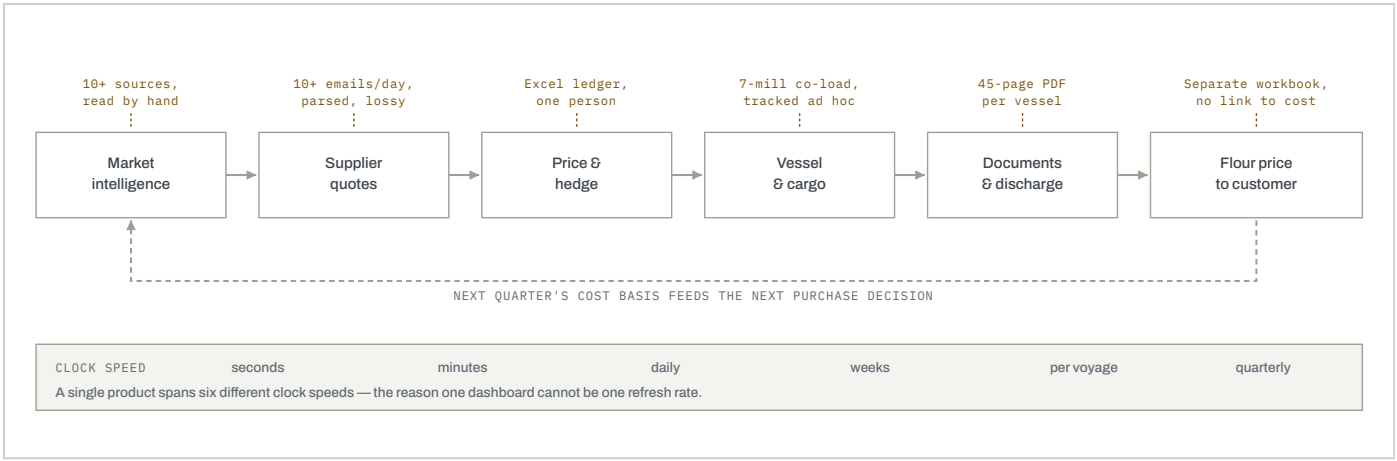
WHERE	WHAT IS ACTUALLY ON SCREEN	WHAT IT CHANGES
18:42	**wheat_price_template** on SharePoint — the Power Automate + Claude Sonnet output. Columns: `date_received · supplier · origin · wheat_class · shipment_period · cnf_flat · cnf_basis · cnf_futures · fob_flat · fob_basis · fob_futures · qty_mt · protein_spec · spec_notes`	**cnf_basis and cnf_futures are nearly always empty** — 2 of ~25 rows. Suppliers quote a flat number. The four-part decomposition is something he must reconstruct, not something he receives.
53:19	**PROCUREMENT TRACKER** — one tab per cargo, plus `Competitor Prices` and `Procurement Notes` tabs. Vessel Pacific Wind, origin PNW, Mitsui, contract 04-Jun-26.	Each cargo is split **70% flat / 30% basis**, and the basis portion is legged into futures over weeks with Open/Filled orders against a take-profit of +$50,000 and a loss cut of −$25,000. Nothing in the notes hinted at this.
17:52	**Q426 Industrial Margin Analysis Pricing** — customers × SKU, ฿/kg ex-VAT, margin %, and forward cost for four quarters. A `Seller` column flips between UFM and MCF.	The selling side exists and is quarterly. Wheat cost becomes a customer price here. A product that stops at the purchase stops one step short of the decision.
22:28	**Presentation to Mitsui:** US ~3–4 ships/yr at ~55K mt; Australian ~3/yr at ~20K; Canadian ~3/yr at ~15K — the latter two **combined across 7 mills**.	~9–10 cargoes a year, not 10–12. Supramax, not uniformly Panamax. And pooling is the **normal structure**, not a fallback — UFM holds 20–40% of those cargoes.
40:32	**(FINAL) DOCS_MV PACIFIC WIND_UFM.pdf** — 45 pages: Baltimore Form C bill of lading, USDA phytosanitary certificate, commodity inspection, SGS.	Paperwork arrives as **one merged pack per vessel**, with certificates issued **per hold**. The job is splitting and reconciling pages inside a PDF, not filing attachments.

> THE ONE THAT SHOULD WORRY US
>
> The Mitsui email at `20:14` quotes five classes. The spreadsheet rows it produced captured four — `NS14.0 = USD373.80` was dropped. His existing automation already loses data silently, and he has no way to know. That is the strongest argument for an intake agent that reports what it could not read instead of quietly writing a row.

THE SHAPE OF THE PROBLEM

One loop, broken in six places

Procurement at UFM is a single loop that runs about ten times a year. Each break in it is a place where Pavin currently substitutes his own attention for a system.



The loop, with the six places it currently breaks. **The clock speeds are the design constraint:** a WASDE release is worth minutes, a quote is worth hours, a document pack is worth a voyage. Any surface that refreshes everything at one rate is wrong for five of the six.

NEEDS HE NAMED

What he asked for, and what answers it

Seven needs came directly from Pavin. Two more we are proposing, marked as ours — he did not ask for them, and they should be presented to him as proposals rather than requirements. All nine are delivered by three agents, set out in the next section.

	WHAT HE SAID	WHAT WE BUILD	WHY IT BITES
N1	*"It's just back and forth emails to negotiate on the prices... I'm shocked it's all done on email." [17:33]*	**Quote Board.** Ingest from the `Price Indications` Outlook folder that already exists (125 items), plus WhatsApp. One email fans out to ~7 rows, one per class. Supersede chains for corrections and recalls. Staleness clock. Comparison on a laid-in basis, never on headline flat.	His own flow already drops quotes silently.
N2	*"That buys me literally five precious minutes to go influence the price... before everyone else reads it." [43:00]*	**Night Watch.** Pre-release expectation sheet, release-second polling, a verdict per class within seconds, the impact priced per cargo, a push to his phone and a pre-drafted message to Mitsui.	He physically stays awake for midnight Bangkok.
N3	*Futures tracked in a workbook he wrote himself; the head of procurement keeps it in his head.*	**Position Book.** Models his sheet exactly: flat/basis split per cargo, contract legs with Open/Filled status, mark-to-market against previous close, breakeven, and the take-profit / loss-cut thresholds he already sets.	Key-person risk on a $19M position.
N4	*"This is all the paperwork that comes in. This isn't the fun part." [46:40]*	**Cargo File.** Split the 45-page vessel pack, classify each page, reconcile per-hold quantities and quality certs against the GAFTA contract spec, and flag what is missing before the vessel arrives.	Discrepancies become claims, and claims expire.

	WHAT HE SAID	WHAT WE BUILD	WHY IT BITES
N5	*Weather, export demand, policy, FX, geopolitics — and a ~$1,500/month consultancy for a weekly outlook.*	**Daily Brief.** Built on *his* causal model from OneNote — weather → crop condition → production → supply, against seasonal and export demand, with Commitment of Traders, government policy and forex acting on price. Each driver carries a direction, a confidence and a $/t sensitivity.	He already reads a bearish/bullish deck daily.
N6	*"Their main thing is they want to help negotiate for you — be the middleman between me and Mitsui… if they save $3 they take 50 cents." [36:11]*	**Negotiator.** Reconstruct the four components, estimate the margin embedded in a flat quote, draft the counter with its justification, remember every concession per supplier, and log the $/t saved.	This is the revenue model, see below.
N7	*Buying cheaper than the other Thai mills — the `Competitor Prices` tab he maintains by hand.*	**Benchmark.** Peer laid-in cost estimated from Thai customs records, vessel line-ups and the co-load cargoes UFM is already inside. Percentile rank per purchase, and the alert he asked for: *locking this cargo puts you $X/t above peer basis.*	The only KPI he actually named.
N8	***Ours.*** *~3 Australian and ~3 Canadian cargoes a year are shared across 7 mills; UFM holds 20–40%.*	**Co-load view.** Tonnage commitments, partner shares, and the laycan each mill is holding. Explicitly *not* a supplier marketplace, which he ruled out twice: *"these flour mills already have the contacts. I know them by name and had drinks with them… it wouldn't be that much needed."* [62:20] Buying is restricted to an approved supplier list anyway.	Pooling is routine, and invisible today.
N9	***Ours.*** *Five classes across three origins get blended into the flour grades priced in the margin workbook.*	**Blend & margin bridge.** Least-cost blend meeting each grade's protein spec at today's prices, carried through to ฿/kg cost per SKU and the margin on each customer's quarterly offer.	Closes the loop to what the mill sells.

> THE NUMBER THAT SETS OUR PRICE
>
> On his own figures, the consultancy's success fee is `$0.50/t × 55,000 t ≈ $27,500 a cargo` — call it **$275,000 a year** across ten cargoes, against an $18,000 annual retainer. The Buyer is not competing with $1,500 a month. It is competing with a quarter of a million dollars a year in success fees, which is also the ceiling on what this product can credibly charge.

THE PLATFORM

Three agents, one desk state

An earlier draft of this document proposed ten agents. That was a list of capabilities wearing costumes — ten things to build, ten things to explain,

and no obvious answer to "which one do I open?" Three is the real number, because there are only three questions Pavin asks in a working week.

Each agent is specified the same way: a trigger, typed inputs, a typed output, a confidence it is willing to state, and a named human who approves anything that leaves the building. They write into one shared desk state and the surfaces render it. **Nothing contacts a supplier on its own.**

WHAT SHOULD I PAY?

The Buyer

Turns a day of unstructured indications into one defensible counter-offer.

Reads each indication into structured rows — one per class, chained onto any quote it corrects or recalls — and names the fields it could not read rather than guessing. Rebuilds futures, basis and freight from the market so the flat price can be taken apart, and reports the residual nobody itemised. Sets that against what the other eight approved suppliers are showing, what this supplier has conceded before, and what the other Thai mills appear to have paid. Then drafts the counter and the argument for it.

TRIGGER	On message arrival; on request before a counter
EMITS	`quote[] · unread_fields[] · components{futures,basis,freight} · residual_margin · peer_basis_est · counter_price · rationale[] · saved_$/t`
APPROVAL	Drafts only. Pavin sends.

FOLDS IN

Quote Clerk · Fair Value · Negotiator · Scout

WHEN SHOULD I MOVE?

The Watch

Owns the clock — the release, the daily picture, and the position carried against both.

Its sharpest mode is the one that wins the demo: at a WASDE or export-sales release it holds a pre-release expectation sheet, diffs the print against it within seconds, scores each class, prices the move against his actual open position and wakes him only if the surprise clears a threshold he set. The rest of the week it writes the morning brief on his own driver taxonomy, and keeps the position book — the 70/30 flat and basis split, the contract legs with Open and Filled status, mark-to-market, breakeven, and the take-profit and loss-cut levels he already sets by hand.

TRIGGER	Release times, to the second; daily 06:00 ICT; on fill; at settlement
EMITS	`verdict/class · surprise_vs_expected · $/t · $/cargo · driver[] {stance,confidence,$/t} · legs[] · mtm · breakeven · target_breach`
APPROVAL	Alerts freely. Orders never — it tells him to call Mitsui.

FOLDS IN

Night Watch · The Analyst · Position Keeper

DID I GET WHAT I BOUGHT?

The Clerk

Follows the cargo from bill of lading to the baht-per-kilo it becomes.

Splits the 45-page vessel pack into its certificates, classifies each page, and reconciles what the paperwork says against what the contract said — per-hold weights against the bill of lading, protein and falling number against the GAFTA spec — raising a claim when they disagree and flagging missing documents before the vessel berths. Then carries the landed cost through the blend: the cheapest mix of the five classes that still meets each flour grade's spec, priced through to ฿/kg per SKU and the margin on each customer's quarterly offer.

TRIGGER	On document pack; on ETA countdown; on price move
EMITS	`page_class[] · completeness · spec_variance · claim_candidate · blend{class:%} · cost_฿/kg · margin_by_SKU`
APPROVAL	Raises claims for review. Never files one.

FOLDS IN

Cargo Clerk · Blender

THE TENTH ONE WAS NEVER AN AGENT

"Desk Memory" — ask anything across every quote, message, document and decision — does not become a fourth agent, and it does not belong inside any of the three. It is a **property of the shared record**: if all three agents write sourced, timestamped output into one store, asking questions of it is a query, not a colleague. Build the store properly and the feature is free. Build three separate agent memories and no amount of work buys it back.

That is the test for anything else proposed later: if it needs its own trigger and its own judgement, it is an agent. If it only needs to read what the three already wrote, it is a surface.

NON-NEGOTIABLE

Every agent output carries its provenance and is allowed to abstain. Nobody commits $19M on an unsourced number, and a parser that silently guesses is worse than one that says what it could not read — we have the dropped `NS14.0` row to prove it.

SURFACE 1 — THE INTERRUPT

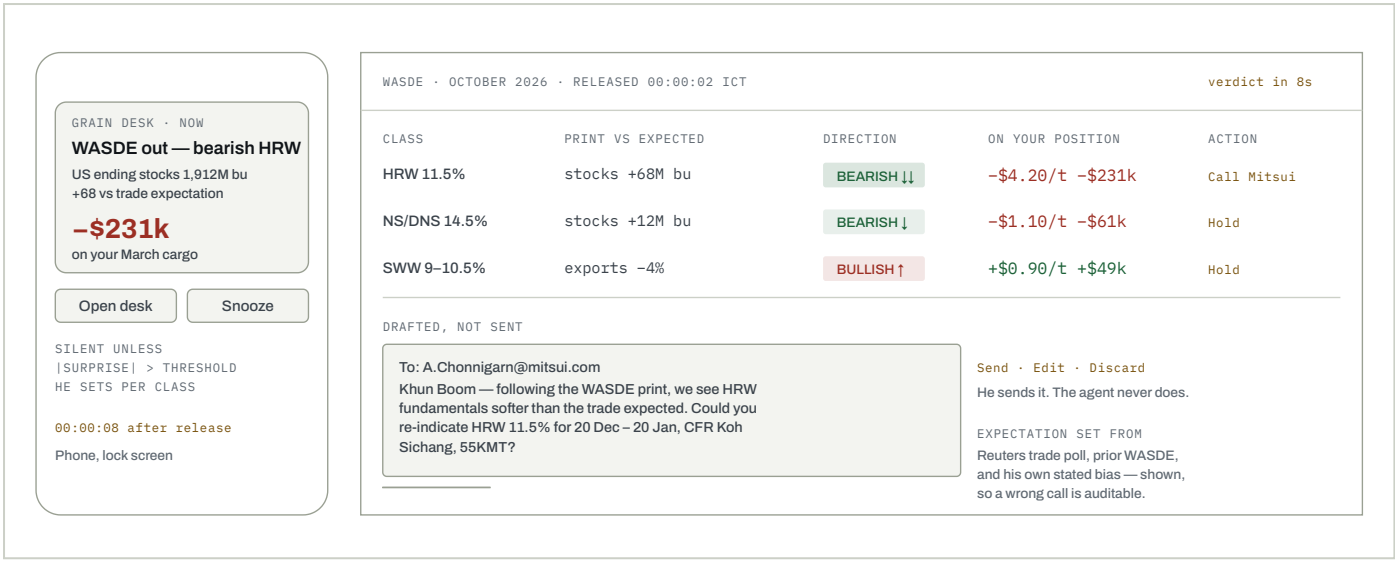
Night Watch

This is not a page. It is an interrupt, designed for a man who is half asleep at midnight and has about five minutes of advantage to spend. It must be readable in one glance, state the consequence in money, and offer exactly one action.

> I literally wait until midnight on these days — or 2am, 1am if there's daylight savings. As soon as it comes out, it's like waiting for your SAT score. You have to fat finger, be very quick to click on this. "Oh, the file's not available yet." … You scroll down to like page 11 or 12, and then — bam. You quickly call Mitsui: if you're in the market right now, buy me this, sell these futures.
>
> Pavin, 41:10 – 42:30

Four things in that passage are specification, not colour. The release lands at **midnight or 2am Bangkok depending on US daylight saving**, so the schedule is not a fixed local time. He is **manually refreshing a page for a PDF that is not there yet**. The wheat numbers he needs are on **page 11 or 12** of a long document. And he already knows the answer:

> You have to have a thing that can instantly — an API that can pull this report out.
>
> Pavin, 42:40

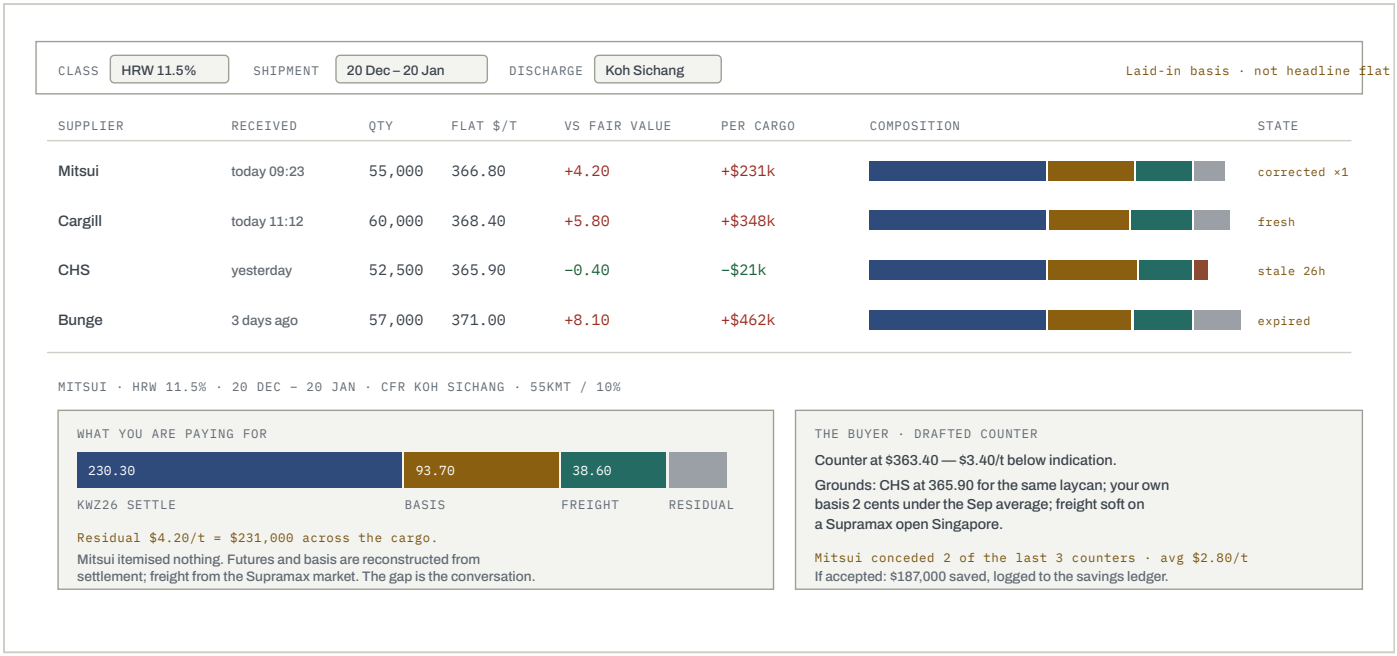


Three rules this wireframe encodes. **Silence is the default** — it fires only when the surprise clears a per-class threshold he sets, or it becomes an alarm he learns to ignore. **The consequence is in money**, against his actual open position, not in bushels. **The message is drafted, never sent** — the agent prepares the five minutes, he spends them.

SURFACE 2 — THE DAILY SURFACE

Quote Board and the decomposition

Ten or more indications a day arrive as flat numbers in prose. The board's job is to make them comparable, and the decomposition's job is to show what is actually inside the number — including the part the supplier did not itemise.



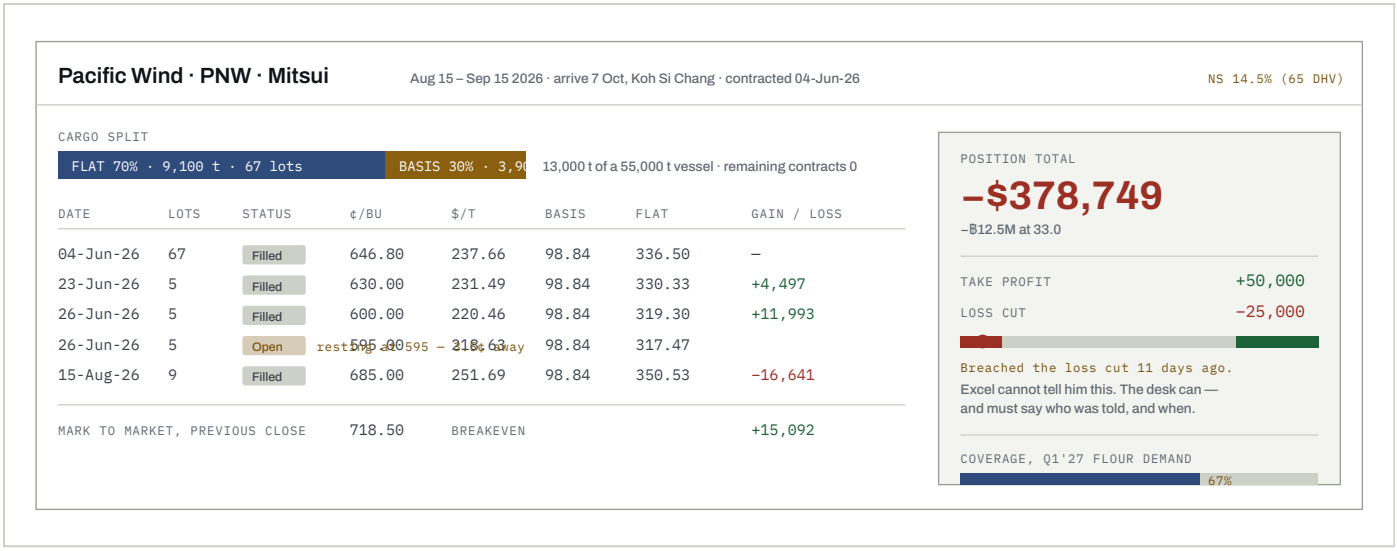
Futures Basis Freight & insurance Margin, itemised Residual — not itemised

The grey block is the point. Because suppliers quote flat and leave `cnf_basis` and `cnf_futures` empty, the honest display is a **reconstruction with its unexplained remainder shown as unexplained** — not a confident four-way split we cannot actually source. That remainder, priced per cargo, is the opening line of the negotiation.

SURFACE 3 — THE POSITION

Position Book

This one we do not get to invent: his workbook already defines it. The design job is to keep his model intact — the flat/basis split, the legs, the thresholds — and add only what Excel cannot do, which is tell him when something has moved.



Everything left of the rail is his spreadsheet, reproduced deliberately — the vocabulary is already his, so we should not rename it. Everything on the rail is what a system adds: **a threshold that was breached eleven days ago and nobody was told.** Coverage against forward flour demand is the one genuinely new column, and it is what turns a trading ledger into a procurement one.

SURFACES 4–8, IN BRIEF

The rest of the desk

Morning Brief — the landing surface

One screen at 07:00 ICT: what moved overnight, which drivers changed stance, what is awaiting his approval, and the one decision today actually requires. Built on his own causal taxonomy — weather, crop condition, production, supply, seasonal, export, demand, Commitment of Traders, government policy, forex — with a bearish/bullish split he already reads daily from his brokers, and a $/t sensitivity on each lever.

Cargo Pipeline

Vessels on the water, each with laycan, discharge port, the co-load partners sharing the hull, and a document completeness meter driven by The Clerk's page classification. Per-hold weights reconciled against the bill of lading; quality certificates against the GAFTA spec. Despatch and

Benchmark & Savings Ledger

Two things on one page because they answer the same question: did we buy well? The benchmark estimates peer laid-in cost with an honest confidence band. The ledger attributes every $/t saved to the action that produced it — a counter accepted, a basis locked on a dip, a freight line challenged. That ledger is what justifies the subscription at renewal.

Blend & margin bridge · Ask the Desk

The bridge carries the purchase through to ฿/kg per flour grade and the margin on each customer's quarterly offer, closing the loop to the workbook at 17 : 52. Ask the Desk is plain retrieval over everything the desk has ever held — the answer to a head of procurement who keeps it in his head.

demurrage exposure sits here too — it appears in his OneNote and in nobody's dashboard.

CAN WE ACTUALLY GET THE DATA

Feasibility, stated honestly

WHAT WE NEED	SOURCE	REALITY
Supplier indications	His `Price Indications` Outlook folder, 125 items, already populated	*Easy. The folder and the routing rule exist. Needs a mailbox decision, not engineering.*
KC HRW & Chicago SRW futures	CME settlements; delayed free, real-time paid	*Easy — but note what he uses today: `"Barchart, it's good enough for me, it's free… it's delayed by 15 minutes, which sucks." [61:50]` Delayed is fine everywhere except Night Watch, where 15 minutes is longer than the entire window of advantage. Real-time at the release moment is precisely the edge he does not currently have.*
MGEX spring wheat `MWU26`	MIAX/MGEX	*Awkward and load-bearing. His biggest class prices off it and there is no free feed. Budget for it.*
WASDE, at the second	USDA ESMIS / NASS	*Feasible. Published to a schedule from a lockup. The engineering is in the pre-release expectation sheet, not the fetch.*
Export sales, COT, crop progress	USDA FAS, CFTC	*Easy, free, weekly.*
Freight	Baltic indices he already reads; Supramax/Panamax time-charter rates, bunkers	*Medium. He was trained by Mitsui to build this from vessel day rates, so the model is known — the rate data is the licence cost.*
FX — USD/THB, AUD, CAD	Any feed	*Trivial.*
Peer cost basis	Thai customs declarations, vessel line-ups, co-load cargoes	***The hard one.*** *Partly inferable, partly not. UFM sits inside 7-mill cargoes, which is real signal — but this may be a data-sharing arrangement between mills rather than a feed, i.e. a business problem wearing an engineering costume. Scheduled last for that reason.*

SEQUENCE

What to build, in what order

The order is a real dependency chain, so the numbering here means something: each stage needs the one before it.

#	STAGE	WHAT IT PROVES, AND WHAT IT NEEDS FIRST
1	**The Watch** — release mode `"Night Watch"`	Wins the demo and the relationship. Standalone — needs only the futures feed and the expectation sheet, no mailbox access, no integration. It is the cheapest thing on this page that he would feel the next time a WASDE lands.

#	STAGE	WHAT IT PROVES, AND WHAT IT NEEDS FIRST
2	**The Buyer** — intake the Quote Board	Replaces the flow he built himself, and fixes its silent data loss. Needs the mailbox decision. Produces the quote history everything downstream depends on.
3	**The Buyer** — fair value and counter	The revenue story — it competes with ~$275k/yr in consultancy success fees. Cannot exist before stage 2, because the counter-offer argument is built from quote history.
4	**The Watch** — position book	Lifts his workbook into shared state and ends the key-person risk. Independent of 2 and 3; can run in parallel if there is capacity.
5	**The Clerk** — document pack	Page splitting and spec reconciliation. Lower urgency, high goodwill — it removes work he openly dislikes.
6	**The Buyer** — benchmark **The Clerk** — blend bridge	Both need accumulated history to be worth anything, and the benchmark needs a data arrangement we do not yet have. Last for good reasons, not as a deprioritisation.

THE RISK PAVIN RAISED HIMSELF

> There might have to be a lot of room for tailoring it to what a company needs — like how Salesforce or SAP would tailor the ERP system to a company based on its needs. They might have their own way of doing things... getting them to adopt the app you guys would make, it has to be very easy, and in their own way that they're comfortable with.
>
> Pavin, 64:10

Our mitigation is that everything specific to a desk — origins, classes, suppliers, instruments, discharge ports, currency, contract standard — lives in a desk profile, not in code. The test of whether that worked is not UFM. It is the second mill, and whether it costs a configuration file or a rewrite.

Note the second half of his warning, which is about adoption rather than architecture: the product has to work *the way he already works*. That is the reason the Position Book reproduces his column names instead of improving them.

> WHERE WE DISAGREE WITH HIM, AND WHY IT IS WORTH RAISING
>
> Asked what to build first, Pavin said: *"the first thing that's easiest to make is the fragmented part, putting things together. The research thing is interesting and that'll be harder."* [65:30]
>
> He is right about difficulty and we are sequencing against him anyway — Night Watch first, consolidation second. The reason is that consolidation is the thing he has *already built for himself*, so rebuilding it proves we can match his spreadsheet; it does not prove we can do something he cannot. Night Watch does, it is standalone, and it lands on a night he is already awake. **This is a disagreement to put to him directly** rather than resolve in a document — he is the one who offered to be a design partner.

BEFORE BUILDING

What we still need to ask him

1. The `Competitor Prices` tab exists — what is in it, and where do those numbers come from? This decides whether N7 is tractable.
2. Is the 70/30 flat/basis split policy, per-cargo negotiation, or his own judgement? It changes whether the Position Book proposes splits or just records them.
3. Who else touches `PROCUREMENT TRACKER` — is it private to him, or shared with Khun Pracha? That determines whether we are replacing a personal tool or a team one, which is a different product.
4. What does AVS actually deliver weekly, and what does the $1,500 buy that his daily broker decks and the US Wheat letters do not?
5. Does the Power Automate flow write only to `wheat_price_template`, or also into the tracker's `Price Indications` tab?
6. Who owns the mailbox decision — and would he pay for the desk, or only for the brokering that comes after it? He deflected this once already.

Two of these are easy to ask, because he volunteered the channel himself: *"You can be a design partner if you'd like... if you have any questions, just schedule a meeting with me."* [66:00]

He also gave us one piece of homework, and it should be done before any of the above: *"Check first to see if there's any tools out there. I doubt it. And maybe that's also — maybe it's because it's too complicated, and that's why there's no tools."* [63:40] That second sentence is the real question. If no one has built this, the interesting possibility is not that nobody noticed;

it is that the per-customer tailoring makes it uneconomic. We should find out which before committing.

Design proposal for discussion. Every quotation and figure is drawn from the 4 October 2026 walkthrough recording and cited by timestamp; screen contents were read from extracted keyframes. Nothing here has been built.